

Numpy Crash Course

```
In [1]: import numpy as np
```

```
In [2]: np.__version__    # To know the version of Numpy in the system
```

```
Out[2]: '2.4.6'
```

```
In [3]: import sys
        sys.version
```

```
Out[3]: '3.14.6 | packaged by Anaconda, Inc. | (main, Jul  9 2026, 14:29:05) [MSC v.194
        2 64 bit (AMD64)]'
```

Creating List

```
In [4]: my_list = [0,1,2,3,4,5]
        my_list
```

```
Out[4]: [0, 1, 2, 3, 4, 5]
```

```
In [5]: type(my_list)
```

```
Out[5]: list
```

Coverision from List to Array

```
In [6]: # !pip install numpy
```

```
In [7]: # How to convert list to array
```

```
arr = np.array(my_list)
arr
```

```
Out[7]: array([0, 1, 2, 3, 4, 5])
```

```
In [8]: type(arr)
```

```
Out[8]: numpy.ndarray
```

```
In [9]: print(type(my_list))
        print(type(arr))
```

```
<class 'list'>
<class 'numpy.ndarray'>
```

```
In [10]: np.    # We Learn important function
```

```
Cell In[10], line 1
      np.    # We learn important function
      ^
SyntaxError: invalid syntax
```

```
In [ ]: # np.arange() -- Create a sequence/range of no.s as a NumPy array

In [11]: np.arange(5) # Numbers from 0 up to, but not including 5. Here, we've given only
Out[11]: array([0, 1, 2, 3, 4])

In [12]: np.arange(5.0) # Gives float values
Out[12]: array([0., 1., 2., 3., 4.])

In [13]: np.arange(9)
Out[13]: array([0, 1, 2, 3, 4, 5, 6, 7, 8])

In [14]: np.arange(5, 10) # Gives no.s from 5 to 9
Out[14]: array([5, 6, 7, 8, 9])

In [15]: np.arange(0, 5)
Out[15]: array([0, 1, 2, 3, 4])

In [16]: np.arange(10, 20)
Out[16]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])

In [17]: np.arange(20, 10) # 1st argument < 2nd argument
Out[17]: array([], dtype=int64)

In [18]: np.arange(5, 100, 10) # Starting point - 5, Ending point - 100 and Step count
Out[18]: array([ 5, 15, 25, 35, 45, 55, 65, 75, 85, 95])

In [19]: np.arange(20, 10)
Out[19]: array([], dtype=int64)

In [20]: np.arange(-20, 10) # Here -20 < 10
Out[20]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
               -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
                6,  7,  8,  9])

In [21]: np.arange(-16, 10)
Out[21]: array([-16, -15, -14, -13, -12, -11, -10, -9, -8, -7, -6, -5, -4,
               -3, -2, -1,  0,  1,  2,  3,  4,  5,  6,  7,  8,  9])

In [22]: np.arange(-20, -10)
Out[22]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11])

In [23]: np.arange(-30, 20)
```

```
Out[23]: array([-30, -29, -28, -27, -26, -25, -24, -23, -22, -21, -20, -19, -18,
              -17, -16, -15, -14, -13, -12, -11, -10, -9, -8, -7, -6, -5,
              -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,  6,  7,  8,
              9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [24]: np.arange(30,20)
```

```
Out[24]: array([], dtype=int64)
```

```
In [25]: np.arange(10, 10) # Both Values are equal, that's why it gives us empty
```

```
Out[25]: array([], dtype=int64)
```

```
In [26]: np.arange(10, 30, 5) # 10 - Starting from 30 - End point, 5 - Step count
```

```
Out[26]: array([10, 15, 20, 25])
```

```
In [27]: np.arange(0, 10, 3)
```

```
Out[27]: array([0, 3, 6, 9])
```

```
In [28]: np.arange(10, 30, 5, 8) # Error - As we've given 4 values
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[28], line 1
----> 1 np.arange(10, 30, 5, 8) # Error - As we've given 4 values

TypeError: Cannot interpret '8' as a data type
```

```
In [ ]: np.arange(10, 100, 10)
```

```
In [ ]: # np.zeros() -- It is a NumPy function used to create an array filled with 0s, h
```

```
In [29]: np.zeros(5) # Parameter tuning (No changing of default of data type value)
```

```
Out[29]: array([0., 0., 0., 0., 0.])
```

```
In [30]: np.zeros(5, dtype=int) # Hyperparameter tuning (Change in default data type val
```

```
Out[30]: array([0, 0, 0, 0, 0])
```

```
In [31]: np.zeros((2,2)) # 2 x 2 Matrix (2 Rows and 2 Columns)
```

```
Out[31]: array([[0., 0.],
               [0., 0.]])
```

```
In [32]: np.zeros((2,2), dtype = int)
```

```
Out[32]: array([[0, 0],
               [0, 0]])
```

```
In [33]: np.zeros((2,3), dtype = int) # 2*3 Matrix (2 Rows and 3 Columns)
```

```
Out[33]: array([[0, 0, 0],
               [0, 0, 0]])
```

```
In [34]: np.zeros((10,3), dtype = int) # 10 * 3 Matrix
```

```
Out[34]: array([[0, 0, 0],
                [0, 0, 0],
                [0, 0, 0],
                [0, 0, 0],
                [0, 0, 0],
                [0, 0, 0],
                [0, 0, 0],
                [0, 0, 0],
                [0, 0, 0],
                [0, 0, 0]])
```

```
In [35]: np.zeros((10,10), dtype=int)
```

```
Out[35]: array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

```
In [36]: zero = np.zeros([2,2])
          print(zero)
          print(type(zero))
```

```
[[0. 0.]
 [0. 0.]]
<class 'numpy.ndarray'>
```

```
In [37]: zero = np.zeros([2,2])
          print(zero)

          print('####')

          print(type(zero))
```

```
[[0. 0.]
 [0. 0.]]
####
<class 'numpy.ndarray'>
```

```
In [38]: np.zeros((2, 10))
```

```
Out[38]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
In [39]: np.zeros((3,3))
```

```
Out[39]: array([[0., 0., 0.],
                [0., 0., 0.],
                [0., 0., 0.]])
```

```
In [40]: np.zeros((10, 30))
```

[illegible]

```
In [41]: np.zeros((10,10), dtype = int)
```

[illegible]

```
In [42]: np.zeros((5,10)) # Bydefault - 1st one is Rows and 2nd one is Columns
```

```
Out[42]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
                [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
In [43]: np.zeros((10,5))
```

[illegible]

```
In [44]: n = (6,7)
          n1 = (6,8)
          print(np.zeros(n1))    # Parameter Tuning
```

```
[[0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]]
```

```
In [45]: print(np.zeros(n, dtype = int))    # Hyperparameter Tunning
```

```
[[0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]]
```

```
In [46]: n
```

```
Out[46]: (6, 7)
```

```
In [47]: n1
```

```
Out[47]: (6, 8)
```

```
In [48]: print(np.zeros(n))
```

```
[[0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]]
```

```
In [ ]: # np.ones() -- It is a NumPy function used to create an array filled with 1s, ha
```

```
In [49]: np.ones(4)    # Parameter tuning
```

```
Out[49]: array([1., 1., 1., 1.])
```

```
In [50]: np.ones(3)
```

```
Out[50]: array([1., 1., 1.])
```

```
In [51]: np.ones((3,3))
```

```
Out[51]: array([[1., 1., 1.],
                [1., 1., 1.],
                [1., 1., 1.]])
```

```
In [52]: np.ones((2,4))
```

```
Out[52]: array([[1., 1., 1., 1.],
                [1., 1., 1., 1.]])
```

```
In [53]: np.ones((6,10), dtype = int)    # Hyperparameter tuning
```

```
Out[53]: array([[1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
                [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
                [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
                [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
                [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
                [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]])
```

```
In [54]: np.ones((2,3), dtype = int)
```

```
Out[54]: array([[1, 1, 1],
                [1, 1, 1]])
```

```
In [55]: np.ones(4, dtype = int)
```

```
Out[55]: array([1, 1, 1, 1])
```

```
In [56]: np.ones(4)
```

```
Out[56]: array([1., 1., 1., 1.])
```

```
In [57]: n
```

```
Out[57]: (6, 7)
```

```
In [58]: np.ones(n)
```

```
Out[58]: array([[1., 1., 1., 1., 1., 1., 1.],
                [1., 1., 1., 1., 1., 1., 1.],
                [1., 1., 1., 1., 1., 1., 1.],
                [1., 1., 1., 1., 1., 1., 1.],
                [1., 1., 1., 1., 1., 1., 1.],
                [1., 1., 1., 1., 1., 1., 1.]])
```

```
In [59]: np.ones((5,4), dtype = int) # Bydefault 5 - Rows and 4 - Columns
```

```
Out[59]: array([[1, 1, 1, 1],
                [1, 1, 1, 1],
                [1, 1, 1, 1],
                [1, 1, 1, 1],
                [1, 1, 1, 1]])
```

```
In [60]: np.twos((2,3)) # There is no two() in numpy exist.
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[60], line 1
----> 1 np.twos((2,3)) # There is no two() in numpy exist.

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:792, in __getattr__(attr)
    789     import numpy.char as char
    790     return char.chararray
--> 792 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'
```

```
In [61]: np.twos((2,4))
```

```

-----
AttributeError                                Traceback (most recent call last)
Cell In[61], line 1
----> 1 np.twos((2,4))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:792, in __getattr__(attr)
    789     import numpy.char as char
    790     return char.chararray
--> 792 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'

```

In [62]: `np.three((2,3))` *# There is no three() in numpy exist.*

```

-----
AttributeError                                Traceback (most recent call last)
Cell In[62], line 1
----> 1 np.three((2,3))  # There is no three() in numpy exist.

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:792, in __getattr__(attr)
    789     import numpy.char as char
    790     return char.chararray
--> 792 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'three'

```

In [63]: `np.three((2,4))`

```

-----
AttributeError                                Traceback (most recent call last)
Cell In[63], line 1
----> 1 np.three((2,4))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:792, in __getattr__(attr)
    789     import numpy.char as char
    790     return char.chararray
--> 792 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'three'

```

In [64]: `range(5)`

Out[64]: `range(0, 5)`

In [65]: `r = range(5)`
`r`

Out[65]: `range(0, 5)`

In [66]: `for i in r:`
 `print(i)`

```

0
1
2
3
4

```

In [67]: `list(range(5))`

Out[67]: [0, 1, 2, 3, 4]

In [68]: `range(1,10)`

Out[68]: `range(1, 10)`

In [69]: `list(range(1,10))`

Out[69]: [1, 2, 3, 4, 5, 6, 7, 8, 9]

In [70]: `list(range(1, 10, 3))`

Out[70]: [1, 4, 7]

In [71]: `y = list(range(12))`
`y`

Out[71]: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]

In [72]: `rand(2,3)`

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[72], line 1  
----> 1 rand(2,3)  
  
NameError: name 'rand' is not defined
```

In [73]: `random.rand(2,3)`

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[73], line 1  
----> 1 random.rand(2,3)  
  
NameError: name 'random' is not defined
```

In []: *# np.random.rand() -- Used to create an array of random numbers between 0 and 1*

In [74]: `np.random.rand(2)`

Out[74]: `array([0.18876816, 0.56239754])`

In [75]: `np.random.rand(5)`

Out[75]: `array([0.75076628, 0.55020802, 0.46940182, 0.57865178, 0.50664581])`

In [76]: `np.rand(4)`

```

-----
AttributeError                                Traceback (most recent call last)
Cell In[76], line 1
----> 1 np.rand(4)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:792, in __getattr__(attr)
    789     import numpy.char as char
    790     return char.chararray
--> 792 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'rand'

```

```
In [77]: np.random.rand(2,3) # Gives random no.s in 2 - Rows and 3 - Columns
```

```
Out[77]: array([[0.42787857, 0.07397896, 0.5515782 ],
                [0.33037341, 0.60688972, 0.98115105]])
```

```
In [78]: np.random.rand(3,5)
```

```
Out[78]: array([[0.62437182, 0.25896519, 0.09177944, 0.90574263, 0.69242802],
                [0.76611498, 0.52813471, 0.78649648, 0.41618294, 0.04037471],
                [0.77166583, 0.0203734 , 0.32173817, 0.56620919, 0.146659  ]])
```

```
In [ ]: # np.random.randint(2) -- Gives random integers
```

```
In [79]: np.random.randint(2) # Generates one random integer from 0 up to 2, but not inc
```

```
Out[79]: 0
```

```
In [80]: np.random.randint(2)
```

```
Out[80]: 1
```

```
In [81]: np.random.rand(3)
```

```
Out[81]: array([0.62691452, 0.73283061, 0.37410956])
```

```
In [82]: np.random.randint(1)
```

```
Out[82]: 0
```

```
In [83]: np.random.rand(3,5)
```

```
Out[83]: array([[0.34554824, 0.77045642, 0.94030587, 0.29287806, 0.60033847],
                [0.9465795 , 0.08109725, 0.79377003, 0.14090125, 0.99435569],
                [0.57612344, 0.77089036, 0.23493921, 0.35111232, 0.11550472]])
```

```
In [84]: np.random.randint(2,9) # Generate no. between 2 to 9
```

```
Out[84]: 8
```

```
In [85]: np.random.randint(4,6)
```

```
Out[85]: 5
```

```
In [ ]:
```

```
In [86]: np.random.randint(2,9,4) # Generate 4 random integers between 2 up to 9 where
```

```
Out[86]: array([3, 6, 8, 3], dtype=int32)
```

```
In [87]: np.random.randint(2,20,4)
```

```
Out[87]: array([ 5, 17,  3, 17], dtype=int32)
```

```
In [88]: np.random.randint(1,6,4)
```

```
Out[88]: array([4, 1, 3, 4], dtype=int32)
```

```
In [89]: np.random.randint(30,20,10) # Can't generate random no.s from 30 to 20. 1st no.
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[89], line 1  
----> 1 np.random.randint(30,20,10)  
  
File numpy/random/mtrand.pyx:799, in numpy.random.mtrand.RandomState.randint()  
--> 799 'Could not get source, probably due dynamically evaluated source code.'  
  
File numpy/random/_bounded_integers.pyx:1425, in numpy.random._bounded_integers._rand_int32()  
-> 1425 'Could not get source, probably due dynamically evaluated source code.'  
  
ValueError: low >= high
```

```
In [90]: np.random.randint(-30,20,10)
```

```
Out[90]: array([ 4, -21, 18,  1, -26,  5, -13, -8, 19,  2], dtype=int32)
```

```
In [91]: np.random.randint(20,30,10)
```

```
Out[91]: array([29, 26, 26, 21, 23, 21, 20, 25, 26, 22], dtype=int32)
```

```
In [92]: np.random.randint(5, 9) # Get the value <=1 & >=5
```

```
Out[92]: 7
```

```
In [93]: np.random.randint(10,21,3)
```

```
Out[93]: array([19, 18, 11], dtype=int32)
```

```
In [94]: np.random.randint(1,12,10)
```

```
Out[94]: array([ 1, 10,  5, 10,  5,  3, 11,  9,  9,  5], dtype=int32)
```

```
In [95]: np.random.randint(10,40,(10,10)) #10*10 matrix between 10 to 40
```

```
Out[95]: array([[35, 32, 10, 12, 16, 36, 11, 11, 22, 15],
               [28, 39, 33, 34, 24, 13, 31, 16, 25, 28],
               [24, 18, 12, 30, 35, 10, 26, 33, 19, 14],
               [13, 10, 22, 31, 29, 18, 31, 24, 29, 26],
               [23, 12, 39, 26, 14, 13, 26, 36, 29, 25],
               [19, 13, 29, 33, 31, 38, 34, 28, 18, 26],
               [17, 17, 13, 25, 28, 30, 26, 10, 33, 10],
               [19, 39, 14, 15, 35, 14, 32, 26, 28, 12],
               [21, 14, 20, 19, 10, 27, 34, 27, 18, 12],
               [32, 37, 11, 16, 36, 16, 29, 30, 33, 19]], dtype=int32)
```

```
In [96]: arr1= np.random.randint(10,40,(10,10))
         print(arr1)
```

```
[[18 37 28 20 24 11 36 20 18 14]
 [32 17 21 39 23 31 17 14 37 31]
 [34 17 11 38 18 36 35 14 20 12]
 [20 10 23 32 28 34 18 20 26 34]
 [12 27 30 24 34 16 39 36 37 38]
 [11 35 34 29 21 32 18 20 24 32]
 [19 26 14 28 16 10 30 38 32 25]
 [24 26 14 11 22 25 26 27 17 34]
 [11 15 29 39 32 25 15 28 24 25]
 [31 27 32 32 23 18 11 10 27 16]]
```

```
In [97]: print(arr1)
```

```
[[18 37 28 20 24 11 36 20 18 14]
 [32 17 21 39 23 31 17 14 37 31]
 [34 17 11 38 18 36 35 14 20 12]
 [20 10 23 32 28 34 18 20 26 34]
 [12 27 30 24 34 16 39 36 37 38]
 [11 35 34 29 21 32 18 20 24 32]
 [19 26 14 28 16 10 30 38 32 25]
 [24 26 14 11 22 25 26 27 17 34]
 [11 15 29 39 32 25 15 28 24 25]
 [31 27 32 32 23 18 11 10 27 16]]
```

```
In [98]: arr1[5]
```

```
Out[98]: array([11, 35, 34, 29, 21, 32, 18, 20, 24, 32], dtype=int32)
```

```
In [99]: np.random.randint(1,100,(12,12))    # 12*12 Matrix between 1 to 100
```

```
Out[99]: array([[98, 62, 26, 60, 90, 64, 86, 85, 61, 57, 61, 41],
               [ 3, 98, 59, 70, 93, 12, 76, 29, 20, 34, 42, 52],
               [72, 62, 63, 63, 17, 97, 72, 99, 51, 73, 83, 38],
               [98, 95, 91, 24, 39, 69, 96, 10, 33,  7, 81, 17],
               [15, 50, 13, 49,  8, 75, 94, 56, 41, 12, 71, 81],
               [ 3, 49, 75, 97, 16, 64, 25, 41, 27, 79, 73, 19],
               [ 5, 73, 74, 38, 12, 50, 63, 82, 21, 44, 51, 51],
               [54, 12, 10, 62,  6, 87, 59, 75, 46, 85, 76, 23],
               [87, 13, 58, 55, 23, 86, 13, 54, 63, 43, 63, 88],
               [90, 64, 43, 47, 28, 49, 94, 91, 90, 69, 57, 62],
               [17,  8, 41, 93, 18, 71, 76, 12, 42, 32, 45, 60],
               [ 8, 41, 41, 88,  4, 51, 28, 82, 51, 78, 37, 22]], dtype=int32)
```

```
In [ ]: # reshape() -- Used to change the shape/ dimensions of an array without changing
```

```
In [2]: a1 = np.arange(6)
        print(a1)
```

```
[0 1 2 3 4 5]
```

```
In [3]: a1.reshape(2,3)
```

```
Out[3]: array([[0, 1, 2],
               [3, 4, 5]])
```

```
In [100... np.arange(1,13)
```

```
Out[100... array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12])
```

```
In [101... np.arange(1,13).reshape(3,4)    # 3*4 Matrix from 1 to 12 no.s, 13 is not incluad
```

```
Out[101... array([[ 1,  2,  3,  4],
                  [ 5,  6,  7,  8],
                  [ 9, 10, 11, 12]])
```

```
In [102... np.arange(1,13).reshape(4,3)
```

```
Out[102... array([[ 1,  2,  3],
                  [ 4,  5,  6],
                  [ 7,  8,  9],
                  [10, 11, 12]])
```

```
In [103... np.arange(1,13).reshape(5,4)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[103], line 1
----> 1 np.arange(1,13).reshape(5,4)

ValueError: cannot reshape array of size 12 into shape (5,4)
```

```
In [104... np.arange(1,13).reshape(6,2)
```

```
Out[104... array([[ 1,  2],
                  [ 3,  4],
                  [ 5,  6],
                  [ 7,  8],
                  [ 9, 10],
                  [11, 12]])
```

```
In [105... np.arange(1,13).reshape(12,1)
```

```
Out[105... array([[ 1],
                  [ 2],
                  [ 3],
                  [ 4],
                  [ 5],
                  [ 6],
                  [ 7],
                  [ 8],
                  [ 9],
                  [10],
                  [11],
                  [12]])
```

Slicing in Matrix

```
In [108... b = np.random.randint(10,20,(5,4))  
b
```

```
Out[108... array([[17, 10, 18, 16],  
        [19, 10, 11, 15],  
        [14, 11, 17, 13],  
        [13, 16, 10, 18],  
        [16, 12, 19, 12]], dtype=int32)
```

```
In [110... np.random.randint(10,20,(5,4))
```

```
Out[110... array([[12, 11, 17, 18],  
        [10, 18, 17, 14],  
        [14, 16, 13, 12],  
        [13, 14, 10, 16],  
        [10, 11, 19, 15]], dtype=int32)
```

```
In [111... type(b)
```

```
Out[111... numpy.ndarray
```

```
In [112... b[:] # Gives all elements
```

```
Out[112... array([[17, 10, 18, 16],  
        [19, 10, 11, 15],  
        [14, 11, 17, 13],  
        [13, 16, 10, 18],  
        [16, 12, 19, 12]], dtype=int32)
```

```
In [113... b[1:3] # Gives output from 1st row up to 3rd row, but 3rd row is excluded
```

```
Out[113... array([[19, 10, 11, 15],  
        [14, 11, 17, 13]], dtype=int32)
```

```
In [114... b[1,2] # Shows the element which is present in 1st index of Row, means 2nd Row
```

```
Out[114... np.int32(11)
```

```
In [115... b
```

```
Out[115... array([[17, 10, 18, 16],  
        [19, 10, 11, 15],  
        [14, 11, 17, 13],  
        [13, 16, 10, 18],  
        [16, 12, 19, 12]], dtype=int32)
```

```
In [116... b[1,3]
```

```
Out[116... np.int32(15)
```

```
In [117... b[1,-1] # '-1'-- Last column (Reverse indexing)
```

```
Out[117... np.int32(15)
```

```
In [118... b
```

```
Out[118...] array([[17, 10, 18, 16],  
          [19, 10, 11, 15],  
          [14, 11, 17, 13],  
          [13, 16, 10, 18],  
          [16, 12, 19, 12]], dtype=int32)
```

```
In [119...] b[2:3]
```

```
Out[119...] array([[14, 11, 17, 13]], dtype=int32)
```

```
In [120...] b
```

```
Out[120...] array([[17, 10, 18, 16],  
          [19, 10, 11, 15],  
          [14, 11, 17, 13],  
          [13, 16, 10, 18],  
          [16, 12, 19, 12]], dtype=int32)
```

```
In [121...] b[0:-2]
```

```
Out[121...] array([[17, 10, 18, 16],  
          [19, 10, 11, 15],  
          [14, 11, 17, 13]], dtype=int32)
```

```
In [123...] b
```

```
Out[123...] array([[17, 10, 18, 16],  
          [19, 10, 11, 15],  
          [14, 11, 17, 13],  
          [13, 16, 10, 18],  
          [16, 12, 19, 12]], dtype=int32)
```

```
In [122...] b[0,2]
```

```
Out[122...] np.int32(18)
```

```
In [124...] b[-5,3]
```

```
Out[124...] np.int32(16)
```

```
In [125...] b[-5,-3]
```

```
Out[125...] np.int32(10)
```

```
In [126...] b
```

```
Out[126...] array([[17, 10, 18, 16],  
          [19, 10, 11, 15],  
          [14, 11, 17, 13],  
          [13, 16, 10, 18],  
          [16, 12, 19, 12]], dtype=int32)
```

```
In [128...] b[-4,-2]
```

```
Out[128...] np.int32(11)
```

```
In [129...] b[-4,2]
```

```
Out[129...] np.int32(11)
```

```
In [130...] b
```

```
Out[130...] array([[17, 10, 18, 16],
                [19, 10, 11, 15],
                [14, 11, 17, 13],
                [13, 16, 10, 18],
                [16, 12, 19, 12]], dtype=int32)
```

```
In [131...] b[:]
```

```
Out[131...] array([[17, 10, 18, 16],
                [19, 10, 11, 15],
                [14, 11, 17, 13],
                [13, 16, 10, 18],
                [16, 12, 19, 12]], dtype=int32)
```

Operations

```
In [133...] a = np.random.randint(10,20,10)
a
```

```
Out[133...] array([13, 10, 11, 17, 19, 18, 10, 16, 15, 15], dtype=int32)
```

```
In [134...] id(a)
```

```
Out[134...] 2395589210736
```

```
In [135...] arr
```

```
Out[135...] array([0, 1, 2, 3, 4, 5])
```

```
In [136...] arr2 = np.random.randint(0,100,(10,10))
```

```
In [137...] arr2
```

```
Out[137...] array([[55,  7, 12, 23, 88, 27, 64, 41, 73, 84],
                [67, 18, 20, 72,  9, 12, 42, 72, 96, 29],
                [31, 91, 87, 56, 44,  8,  5, 19, 76,  9],
                [31, 50, 46, 20, 49, 70, 71, 13,  2, 60],
                [80, 26,  2, 17,  7, 21, 41, 99, 71, 72],
                [37,  6, 77,  2, 26, 54, 83, 94, 69, 45],
                [16, 59,  0, 53, 62, 97, 15, 56, 32, 80],
                [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
                [36, 39, 17, 13,  4, 25, 97, 45, 31, 57],
                [57, 55, 93, 50, 77, 31, 62, 67,  4, 53]], dtype=int32)
```

```
In [138...] arr
```

```
Out[138...] array([0, 1, 2, 3, 4, 5])
```

```
In [139...] arr[:]
```

```
Out[139...] array([0, 1, 2, 3, 4, 5])
```

```
In [140... arr
```

```
Out[140... array([0, 1, 2, 3, 4, 5])
```

```
In [141... arr[:4]
```

```
Out[141... array([0, 1, 2, 3])
```

```
In [142... arr2[:]
```

```
Out[142... array([[55,  7, 12, 23, 88, 27, 64, 41, 73, 84],
        [67, 18, 20, 72,  9, 12, 42, 72, 96, 29],
        [31, 91, 87, 56, 44,  8,  5, 19, 76,  9],
        [31, 50, 46, 20, 49, 70, 71, 13,  2, 60],
        [80, 26,  2, 17,  7, 21, 41, 99, 71, 72],
        [37,  6, 77,  2, 26, 54, 83, 94, 69, 45],
        [16, 59,  0, 53, 62, 97, 15, 56, 32, 80],
        [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
        [36, 39, 17, 13,  4, 25, 97, 45, 31, 57],
        [57, 55, 93, 50, 77, 31, 62, 67,  4, 53]]], dtype=int32)
```

```
In [143... arr2[0:5]
```

```
Out[143... array([[55,  7, 12, 23, 88, 27, 64, 41, 73, 84],
        [67, 18, 20, 72,  9, 12, 42, 72, 96, 29],
        [31, 91, 87, 56, 44,  8,  5, 19, 76,  9],
        [31, 50, 46, 20, 49, 70, 71, 13,  2, 60],
        [80, 26,  2, 17,  7, 21, 41, 99, 71, 72]]], dtype=int32)
```

```
In [144... arr2
```

```
Out[144... array([[55,  7, 12, 23, 88, 27, 64, 41, 73, 84],
        [67, 18, 20, 72,  9, 12, 42, 72, 96, 29],
        [31, 91, 87, 56, 44,  8,  5, 19, 76,  9],
        [31, 50, 46, 20, 49, 70, 71, 13,  2, 60],
        [80, 26,  2, 17,  7, 21, 41, 99, 71, 72],
        [37,  6, 77,  2, 26, 54, 83, 94, 69, 45],
        [16, 59,  0, 53, 62, 97, 15, 56, 32, 80],
        [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
        [36, 39, 17, 13,  4, 25, 97, 45, 31, 57],
        [57, 55, 93, 50, 77, 31, 62, 67,  4, 53]]], dtype=int32)
```

```
In [145... arr2[1,4]
```

```
Out[145... np.int32(9)
```

```
In [146... arr2
```

```
Out[146... array([[55,  7, 12, 23, 88, 27, 64, 41, 73, 84],
        [67, 18, 20, 72,  9, 12, 42, 72, 96, 29],
        [31, 91, 87, 56, 44,  8,  5, 19, 76,  9],
        [31, 50, 46, 20, 49, 70, 71, 13,  2, 60],
        [80, 26,  2, 17,  7, 21, 41, 99, 71, 72],
        [37,  6, 77,  2, 26, 54, 83, 94, 69, 45],
        [16, 59,  0, 53, 62, 97, 15, 56, 32, 80],
        [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
        [36, 39, 17, 13,  4, 25, 97, 45, 31, 57],
        [57, 55, 93, 50, 77, 31, 62, 67,  4, 53]]], dtype=int32)
```

```
In [147... arr2[-5,-5]
```

```
Out[147... np.int32(54)
```

```
In [148... arr2
```

```
Out[148... array([[55,  7, 12, 23, 88, 27, 64, 41, 73, 84],
        [67, 18, 20, 72,  9, 12, 42, 72, 96, 29],
        [31, 91, 87, 56, 44,  8,  5, 19, 76,  9],
        [31, 50, 46, 20, 49, 70, 71, 13,  2, 60],
        [80, 26,  2, 17,  7, 21, 41, 99, 71, 72],
        [37,  6, 77,  2, 26, 54, 83, 94, 69, 45],
        [16, 59,  0, 53, 62, 97, 15, 56, 32, 80],
        [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
        [36, 39, 17, 13,  4, 25, 97, 45, 31, 57],
        [57, 55, 93, 50, 77, 31, 62, 67,  4, 53]]], dtype=int32)
```

```
In [149... arr2[-5,5]
```

```
Out[149... np.int32(54)
```

```
In [150... arr2[-1,-2]
```

```
Out[150... np.int32(4)
```

```
In [151... arr2[::-1] # Reverse the array or Move backwards
```

```
Out[151... array([[57, 55, 93, 50, 77, 31, 62, 67,  4, 53],
        [36, 39, 17, 13,  4, 25, 97, 45, 31, 57],
        [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
        [16, 59,  0, 53, 62, 97, 15, 56, 32, 80],
        [37,  6, 77,  2, 26, 54, 83, 94, 69, 45],
        [80, 26,  2, 17,  7, 21, 41, 99, 71, 72],
        [31, 50, 46, 20, 49, 70, 71, 13,  2, 60],
        [31, 91, 87, 56, 44,  8,  5, 19, 76,  9],
        [67, 18, 20, 72,  9, 12, 42, 72, 96, 29],
        [55,  7, 12, 23, 88, 27, 64, 41, 73, 84]]], dtype=int32)
```

```
In [152... arr2
```

```
Out[152... array([[55,  7, 12, 23, 88, 27, 64, 41, 73, 84],
        [67, 18, 20, 72,  9, 12, 42, 72, 96, 29],
        [31, 91, 87, 56, 44,  8,  5, 19, 76,  9],
        [31, 50, 46, 20, 49, 70, 71, 13,  2, 60],
        [80, 26,  2, 17,  7, 21, 41, 99, 71, 72],
        [37,  6, 77,  2, 26, 54, 83, 94, 69, 45],
        [16, 59,  0, 53, 62, 97, 15, 56, 32, 80],
        [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
        [36, 39, 17, 13,  4, 25, 97, 45, 31, 57],
        [57, 55, 93, 50, 77, 31, 62, 67,  4, 53]]], dtype=int32)
```

```
In [153... arr2[::-2] # Take elements backwards, skipping every alternate elements.
```

```
Out[153... array([[57, 55, 93, 50, 77, 31, 62, 67,  4, 53],
        [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
        [37,  6, 77,  2, 26, 54, 83, 94, 69, 45],
        [31, 50, 46, 20, 49, 70, 71, 13,  2, 60],
        [67, 18, 20, 72,  9, 12, 42, 72, 96, 29]]], dtype=int32)
```

```
In [154... arr2[::-3] # Go backwards with a step of 3.
```

```
Out[154... array([[57, 55, 93, 50, 77, 31, 62, 67, 4, 53],
        [16, 59, 0, 53, 62, 97, 15, 56, 32, 80],
        [31, 50, 46, 20, 49, 70, 71, 13, 2, 60],
        [55, 7, 12, 23, 88, 27, 64, 41, 73, 84]]], dtype=int32)
```

```
In [155... arr2
```

```
Out[155... array([[55, 7, 12, 23, 88, 27, 64, 41, 73, 84],
        [67, 18, 20, 72, 9, 12, 42, 72, 96, 29],
        [31, 91, 87, 56, 44, 8, 5, 19, 76, 9],
        [31, 50, 46, 20, 49, 70, 71, 13, 2, 60],
        [80, 26, 2, 17, 7, 21, 41, 99, 71, 72],
        [37, 6, 77, 2, 26, 54, 83, 94, 69, 45],
        [16, 59, 0, 53, 62, 97, 15, 56, 32, 80],
        [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
        [36, 39, 17, 13, 4, 25, 97, 45, 31, 57],
        [57, 55, 93, 50, 77, 31, 62, 67, 4, 53]]], dtype=int32)
```

```
In [156... arr2[: -3]
```

```
Out[156... array([[55, 7, 12, 23, 88, 27, 64, 41, 73, 84],
        [67, 18, 20, 72, 9, 12, 42, 72, 96, 29],
        [31, 91, 87, 56, 44, 8, 5, 19, 76, 9],
        [31, 50, 46, 20, 49, 70, 71, 13, 2, 60],
        [80, 26, 2, 17, 7, 21, 41, 99, 71, 72],
        [37, 6, 77, 2, 26, 54, 83, 94, 69, 45],
        [16, 59, 0, 53, 62, 97, 15, 56, 32, 80]]], dtype=int32)
```

```
In [157... arr2[0:10:3]
```

```
Out[157... array([[55, 7, 12, 23, 88, 27, 64, 41, 73, 84],
        [31, 50, 46, 20, 49, 70, 71, 13, 2, 60],
        [16, 59, 0, 53, 62, 97, 15, 56, 32, 80],
        [57, 55, 93, 50, 77, 31, 62, 67, 4, 53]]], dtype=int32)
```

```
In [158... arr2
```

```
Out[158... array([[55, 7, 12, 23, 88, 27, 64, 41, 73, 84],
        [67, 18, 20, 72, 9, 12, 42, 72, 96, 29],
        [31, 91, 87, 56, 44, 8, 5, 19, 76, 9],
        [31, 50, 46, 20, 49, 70, 71, 13, 2, 60],
        [80, 26, 2, 17, 7, 21, 41, 99, 71, 72],
        [37, 6, 77, 2, 26, 54, 83, 94, 69, 45],
        [16, 59, 0, 53, 62, 97, 15, 56, 32, 80],
        [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
        [36, 39, 17, 13, 4, 25, 97, 45, 31, 57],
        [57, 55, 93, 50, 77, 31, 62, 67, 4, 53]]], dtype=int32)
```

```
In [159... arr2[3:]
```

```
Out[159... array([[31, 50, 46, 20, 49, 70, 71, 13, 2, 60],
        [80, 26, 2, 17, 7, 21, 41, 99, 71, 72],
        [37, 6, 77, 2, 26, 54, 83, 94, 69, 45],
        [16, 59, 0, 53, 62, 97, 15, 56, 32, 80],
        [86, 23, 40, 60, 23, 15, 52, 80, 59, 29],
        [36, 39, 17, 13, 4, 25, 97, 45, 31, 57],
        [57, 55, 93, 50, 77, 31, 62, 67, 4, 53]]], dtype=int32)
```

```
In [160... arr
```

```
Out[160... array([0, 1, 2, 3, 4, 5])
```

```
In [ ]: # arr.max() -- Used to find the largest value.
```

```
In [161... arr.max()
```

```
Out[161... np.int64(5)
```

```
In [ ]: # arr.min() -- Used to find the smallest value.
```

```
In [162... arr.min()
```

```
Out[162... np.int64(0)
```

```
In [163... arr
```

```
Out[163... array([0, 1, 2, 3, 4, 5])
```

```
In [ ]: # arr.mean() -- Used to find the average(mean) of all values in a NumPy array
```

```
In [164... arr.mean()
```

```
Out[164... np.float64(2.5)
```

```
In [165... arr
```

```
Out[165... array([0, 1, 2, 3, 4, 5])
```

```
In [ ]: # arr.median -- Finds the middle vlaue of an array after arranging the values in
```

```
In [167... arr.median()
```

```
-----  
AttributeError                                Traceback (most recent call last)  
Cell In[167], line 1  
----> 1 arr.median()  
  
AttributeError: 'numpy.ndarray' object has no attribute 'median'
```

```
In [4]: arr1 = [1,2,3,4,5,6,7]  
arr1
```

```
Out[4]: [1, 2, 3, 4, 5, 6, 7]
```

```
In [5]: arr1.median()
```

```
-----  
AttributeError                                Traceback (most recent call last)  
Cell In[5], line 1  
----> 1 arr1.median()  
  
AttributeError: 'list' object has no attribute 'median'
```

```
In [168... from numpy import *      # 256 function called Internally
a = array([1,2,3,4,9])
median(a)
```

```
Out[168... np.float64(3.0)
```

```
In [169... arr
```

```
Out[169... array([0, 1, 2, 3, 4, 5])
```

```
In [170... arr.reshape(3,2)
```

```
Out[170... array([[0, 1],
        [2, 3],
        [4, 5]])
```

```
In [171... arr.reshape(6,1)
```

```
Out[171... array([[0],
        [1],
        [2],
        [3],
        [4],
        [5]])
```

```
In [172... arr.reshape(1,6)
```

```
Out[172... array([[0, 1, 2, 3, 4, 5]])
```

```
In [173... arr
```

```
Out[173... array([0, 1, 2, 3, 4, 5])
```

```
In [175... arr.reshape(2,4)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[175], line 1
----> 1 arr.reshape(2,4)

ValueError: cannot reshape array of size 6 into shape (2,4)
```

```
In [176... arr
```

```
Out[176... array([0, 1, 2, 3, 4, 5])
```

```
In [177... arr.reshape(3,2, order='C')      # 'C' -- Reads the element row by row / row-wise
```

```
Out[177... array([[0, 1],
        [2, 3],
        [4, 5]])
```

```
In [178... arr.reshape(2,3, order='C')
```

```
Out[178... array([[0, 1, 2],
        [3, 4, 5]])
```

```
In [179... arr.reshape(2,3, order='F')      # Print element with Fortran style ('F') -- Fills
```

Out[179... array([[0, 2, 4],
[1, 3, 5]])

In [180... `arr.reshape(2,3, order='A')` # 'A' almost give you C type output ('A' means "A"

Out[180... array([[0, 1, 2],
[3, 4, 5]])

```
In [181... # importing Numpy package
import numpy as np

# Creating a Numpy array
num_array = np.arange(100).reshape(20,5)
print(num_array)

# Display array in Fortran order

print("\n C type array:")
for num_array in np.nditer(num_array, order="C"):
    print(num_array, end=' ')
```

```
[[ 0  1  2  3  4]
 [ 5  6  7  8  9]
 [10 11 12 13 14]
 [15 16 17 18 19]
 [20 21 22 23 24]
 [25 26 27 28 29]
 [30 31 32 33 34]
 [35 36 37 38 39]
 [40 41 42 43 44]
 [45 46 47 48 49]
 [50 51 52 53 54]
 [55 56 57 58 59]
 [60 61 62 63 64]
 [65 66 67 68 69]
 [70 71 72 73 74]
 [75 76 77 78 79]
 [80 81 82 83 84]
 [85 86 87 88 89]
 [90 91 92 93 94]
 [95 96 97 98 99]]
```

C type array:

```
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 3
0 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 5
7 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 8
4 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
```

```
In [182... # importing Numpy package
import numpy as np

# Creating a Numpy array
num_array = np.arange(100).reshape(20,5)
print(num_array)

# Display array in Fortran order

print("\n C type array:")
for num_array in np.nditer(num_array, order="F"):
    print(num_array, end=' ')
```

```

[[ 0  1  2  3  4]
 [ 5  6  7  8  9]
 [10 11 12 13 14]
 [15 16 17 18 19]
 [20 21 22 23 24]
 [25 26 27 28 29]
 [30 31 32 33 34]
 [35 36 37 38 39]
 [40 41 42 43 44]
 [45 46 47 48 49]
 [50 51 52 53 54]
 [55 56 57 58 59]
 [60 61 62 63 64]
 [65 66 67 68 69]
 [70 71 72 73 74]
 [75 76 77 78 79]
 [80 81 82 83 84]
 [85 86 87 88 89]
 [90 91 92 93 94]
 [95 96 97 98 99]]

```

C type array:

```

0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 1 6 11 16 21 26 31 36 4
1 46 51 56 61 66 71 76 81 86 91 96 2 7 12 17 22 27 32 37 42 47 52 57 62 67 72 77
82 87 92 97 3 8 13 18 23 28 33 38 43 48 53 58 63 68 73 78 83 88 93 98 4 9 14 19 2
4 29 34 39 44 49 54 59 64 69 74 79 84 89 94 99

```

In [183... `arr.reshape(2,3)`

Out[183... `array([[0, 1, 2],
[3, 4, 5]])`

In [184... `arr.reshape(1,4)`

```

-----
ValueError                                Traceback (most recent call last)
Cell In[184], line 1
----> 1 arr.reshape(1,4)

ValueError: cannot reshape array of size 6 into shape (1,4)

```

In [185... `arr.reshape(1,6)`

Out[185... `array([[0, 1, 2, 3, 4, 5]])`

In [186... `arr.reshape(6,1)`

Out[186... `array([[0],
[1],
[2],
[3],
[4],
[5]])`

In [187... `arr.reshape(2,6)`

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[187], line 1  
----> 1 arr.reshape(2,6)  
  
ValueError: cannot reshape array of size 6 into shape (2,6)
```

```
In [188... arr.reshape(3,3)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[188], line 1  
----> 1 arr.reshape(3,3)  
  
ValueError: cannot reshape array of size 6 into shape (3,3)
```

```
In [189... arr
```

```
Out[189... array([0, 1, 2, 3, 4, 5])
```

```
In [190... arr.reshape(3,2)
```

```
Out[190... array([[0, 1],  
                [2, 3],  
                [4, 5]])
```

```
In [191... ## Indexing
```

```
In [192... mat = np.arange(0,100).reshape(10,10)
```

```
In [193... mat
```

```
Out[193... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [194... row = 4  
col = 5
```

```
In [195... col
```

```
Out[195... 5
```

```
In [196... row
```

```
Out[196... 4
```

```
In [197... mat
```

```
Out[197...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [198...] mat[row,col]
```

```
Out[198...] np.int64(45)
```

```
In [199...] mat[4,5]
```

```
Out[199...] np.int64(45)
```

```
In [200...] mat
```

```
Out[200...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [201...] mat[:]
```

```
Out[201...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [202...] col = 6
```

```
In [203...] mat
```

```
Out[203...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [204...] mat[7] # Default, it represents to rows
```

```
Out[204...] array([70, 71, 72, 73, 74, 75, 76, 77, 78, 79])
```

```
In [205...] mat
```

```
Out[205...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
 [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
 [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
 [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
 [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
 [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
 [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
 [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
 [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
 [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [206...] # With Slices
mat[:,col]
```

```
Out[206...] array([ 6, 16, 26, 36, 46, 56, 66, 76, 86, 96])
```

```
In [207...] mat
```

```
Out[207...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
 [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
 [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
 [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
 [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
 [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
 [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
 [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
 [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
 [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [208...] mat[:, -1]
```

```
Out[208...] array([ 9, 19, 29, 39, 49, 59, 69, 79, 89, 99])
```

```
In [209...] mat
```

```
Out[209...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
 [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
 [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
 [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
 [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
 [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
 [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
 [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
 [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
 [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [210...] mat[row, :]
```

```
Out[210...] array([40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [211...] mat
```

```
Out[211...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [212...] mat[:,8]
```

```
Out[212...] array([ 8, 18, 28, 38, 48, 58, 68, 78, 88, 98])
```

```
In [213...] mat[:,col]
```

```
Out[213...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [214...] mat
```

```
Out[214...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [216...] mat[:6]
```

```
Out[216...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [217...] row
```

```
Out[217...] 4
```

```
In [218...] mat
```

```
Out[218...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [219...] mat[:row]
```

```
Out[219...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]])
```

```
In [220...] mat
```

```
Out[220...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [221...] mat[row:]
```

```
Out[221...] array([[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [222...] mat[:]
```

```
Out[222...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [223...] mat[:,8]
```

```
Out[223...] array([ 8, 18, 28, 38, 48, 58, 68, 78, 88, 98])
```

```
In [224...] mat
```

```
Out[224...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [225...] mat[:, -1]
```

```
Out[225...] array([ 9, 19, 29, 39, 49, 59, 69, 79, 89, 99])
```

```
In [226...] mat
```

```
Out[226...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [227...] row
```

```
Out[227...] 4
```

```
In [228...] col
```

```
Out[228...] 6
```

```
In [229...] mat[:, col]
```

```
Out[229...] array([ 6, 16, 26, 36, 46, 56, 66, 76, 86, 96])
```

```
In [230...] mat
```

```
Out[230...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [231...] mat[1, 4]
```

```
Out[231...] np.int64(14)
```

```
In [232...] mat[1:4]
```

```
Out[232...] array([[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]])
```

```
In [233...] mat[3:-3]
```

```
Out[233...] array([[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

```
In [234...] mat
```

```
Out[234...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [235...] mat[0]
```

```
Out[235...] array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [236...] mat[6]
```

```
Out[236...] array([60, 61, 62, 63, 64, 65, 66, 67, 68, 69])
```

```
In [237...] mat
```

```
Out[237...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [238...] mat[6:]
```

```
Out[238...] array([[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [239...] mat[:6]
```

```
Out[239...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [240...] mat
```

```
Out[240...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [241...] mat[5:7]
```

```
Out[241...] array([[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

```
In [242...] mat[0:10]
```

```
Out[242...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [243...] mat
```

```
Out[243...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [244...] mat[0:10:3]
```

```
Out[244...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [245...] mat[0:10]
```

```
Out[245...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [246...] mat
```

```
Out[246...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [247...] mat[4:]
```

```
Out[247...] array([[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [248...] mat[:4]
```

```
Out[248...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]])
```

```
In [249...] mat
```

```
Out[249...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [250...] mat[:, :-1]
```

```
Out[250...] array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

```
In [251...] mat[:, :-3]
```

```
Out[251...] array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

```
In [252...] mat[:, :-5]
```

```
Out[252...] array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49]])
```

```
In [253...] mat
```

```
Out[253...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [254...] mat[2:6]
```

```
Out[254...] array([[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [255...] mat
```

```
Out[255...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [256...] mat[2:6, 2:4] # 1:5 --> Only row part /// 1:3 -- It indicates only column parts
```

```
Out[256...] array([[22, 23],  
          [32, 33],  
          [42, 43],  
          [52, 53]])
```

```
In [257...] mat
```

```
Out[257...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [258...] mat[0,1]
```

```
Out[258...] np.int64(1)
```

```
In [259...] mat[1,6]
```

```
Out[259...] np.int64(16)
```

```
In [260...] mat[1:6]
```

```
Out[260...] array([[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [261...] mat
```

```
Out[261...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [262...] mat[1:]
```

```
Out[262...] array([[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [263... `mat[:6]`

Out[263... `array([[0, 1, 2, 3, 4, 5, 6, 7, 8, 9],
 [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
 [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
 [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
 [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
 [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])`

In [264... `mat[0:1]`

Out[264... `array([[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]])`

In [265... `mat`

Out[265... `array([[0, 1, 2, 3, 4, 5, 6, 7, 8, 9],
 [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
 [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
 [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
 [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
 [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
 [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
 [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
 [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
 [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])`

In [266... `mat[3:5]`

Out[266... `array([[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
 [40, 41, 42, 43, 44, 45, 46, 47, 48, 49]])`

In [267... `mat[3,5]`

Out[267... `np.int64(35)`

In [268... `mat`

Out[268... `array([[0, 1, 2, 3, 4, 5, 6, 7, 8, 9],
 [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
 [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
 [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
 [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
 [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
 [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
 [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
 [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
 [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])`

In [269... `mat[1:2,2:4]`

Out[269... `array([[12, 13]])`

In [270... `mat`

```
Out[270...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [271...] mat[2:3,2:3]
```

```
Out[271...] array([[22]])
```

```
In [272...] mat[3:5,2:4]
```

```
Out[272...] array([[32, 33],
        [42, 43]])
```

```
In [273...] mat[2:3,4:5]
```

```
Out[273...] array([[24]])
```

```
In [274...] ## Masking
```

```
In [275...] mat # We also called as filter
```

```
Out[275...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [276...] id(mat)
```

```
Out[276...] 2395643104880
```

```
In [277...] mat > 50
```

```
Out[277...] array([[False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, True, True, True, True, True, True, True, True,
        True],
        [ True, True, True, True, True, True, True, True, True,
        True],
        [ True, True, True, True, True, True, True, True, True,
        True],
        [ True, True, True, True, True, True, True, True, True,
        True],
        [ True, True, True, True, True, True, True, True, True,
        True]])
```

```
In [278...] mat[mat == 50]
```

```
Out[278...] array([50])
```

```
In [279...] mat
```

```
Out[279...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [280...] mat == 50
```

```
Out[280...] array([[False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [ True, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False]])
```

```
In [281...] a1 = mat[mat<50]
a1
```

```
Out[281...] array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
        34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [282...] mat
```

```
Out[282...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [283...] a2 = mat[mat>50]
a2
```

```
Out[283...] array([51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67,
        68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84,
        85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99])
```

```
In [284...] a3 = mat[mat<=50]
a3
```

```
Out[284...] array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
        34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50])
```

```
In [285...] a4 = mat[mat==50]
a4
```

```
Out[285...] array([50])
```

In [286... *## Pyhton Program to Generate OTP*

```
In [288... import random

def generate_otp(length = 4):
    """Generate a numeric OTP of a specified length."""
    digits = '012345'
    otp = ''.join(random.choice(digits) for _ in range(length))
    return otp

# Example usage
otp_length = 4 # You can change this to any length you prefer
otp = generate_otp(otp_length)
print(f"Your OTP is: {otp}")
```

Your OTP is: 5445

In []: